

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 3 – MCQ Test

Course Code:	DSA0304	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	G.V.S.HEMANTH	Reg. No.:	192425186
Section / Batch:	AIML/ SLOT B	Date:	17-08-2026

Code of Conduct

I G.V.S.HEMANTH/192425186 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action

Signature of the Student: G.V.S.HEMANTH

Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Context-Free Grammars (CFG)	CFG components, productions, terminals, non-terminals, recursion, sentence generation	1 – 10
Sentence Construction and Agreement	Parse trees, sentence structure, grammatical agreement, syntactic validation	11 – 20
Subcategorization	Verb frames, complement requirements, argument structure analysis	21 – 25
Top-Down Parsing	Parsing strategy, derivation process, recursion issues, grammar expansion	26 – 30
Earley Parsing	Predictor, Scanner, Completer operations, ambiguity handling	31 – 35
Feature Structures	Feature representation, unification, constraint checking, agreement enforcement	36 – 40
Probabilistic Context-Free Grammars (PCFG)	Rule probabilities, ambiguity resolution, probabilistic parsing	41 – 48
Integrated Syntax Analysis	CFG + Feature Structures + PCFG integration in NLP applications	49 – 50

Annexure A – MCQ Test

<https://forms.gle/WSv28zfaU5B2aWcy6>

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Number of Correct Answers	100%	45 to 50 correct answers	38 to 44 correct answers	25 to 37 correct answers	0 to 24 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

No. of Questions	Number of Correct Answers	Accuracy	Total %
50			

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Name: G. V. S. Hemanth

Roll No: 2590304

Reg No: 192225186

Date: 14/08/24

1A

☒ (D)

2A

☒ (A)

subject-verb agreement

3A

☒ (C)

finite state automation

4A

☒ (A)

morphological analysis

5A

☒ (B)

6A

☒ (B)

uses, predictor, scanner and computer operation to parse partial.

7A

☒ (A)

regular expression matching

8A

☒ (A)

choose the parse tree with the maximum rule probabilities.

9A

☒ (C)

finite state transducer parser scanner

10A

☒ (A)

CFG + feature structure subcategorization PCFG

11A

☒ (B)

remove all PP productions from the grammar

12A

☒ (A)

The list of authorized employees were updated.

13A

☒ (C)

The verb violates its subcategorization frame.

27
30

49) ~~(B)~~ Left recursion causing infinite recursion.

50) ~~(D)~~ Dynamic chart construction combining
predictor, scanner, and completer.

(6A) ~~(C)~~ Number feature conflict between subject
and verb.

(7A) ~~(C)~~ Rec. because it has higher probability.

(8A) ~~(D)~~ High recursive grammars with unrestricted
productions.

10A) ~~(C)~~ Easily parsing ambiguity efficiently while
PCFG works alternative.

20A) ~~(C)~~ CFG parsing $\rightarrow$ subcategorization checking.

21A ~~(C)~~ Introductory a compound noun production $\rightarrow$
"living room".

22A ~~(B)~~ Object noun phrase.

23A ~~(C)~~ Verb subcategorization frame.

24A ~~(B)~~ Lexical ambiguity.

25A ~~(C)~~ Scanner.

26A ~~(C)~~ maximum likelihood estimation.

27A ~~(C)~~ Tree A it has the highest probability.

28A ~~(C)~~ Early predictor.

29A ~~(C)~~ Recursive top-down parser.

30A ~~(C)~~ Feature agreement.

32 ~~(A)~~ S → NPVP → DetNP → the robot

33 ~~(B)~~ The engineer develops the software.

34 ~~(C)~~ she gave the customer a refund.

35 ~~(D)~~ Expand VP.

36 ~~(E)~~ scanner.

37 ~~(F)~~ unification fails because feature values conflict.

38 ~~(G)~~ the parse with the highest overall probability.

39 ~~(H)~~ recursive descent parsing.

40 ~~(I)~~ parse A because it satisfies constraints and the highest probability.

41 ~~(J)~~ reject the parse because grammatical.

42 ~~(K)~~ NP → Det N

43 ~~(L)~~ VP (A) Det

44 ~~(M)~~ complete.

45 ~~(N)~~ sentence accepted because reason matched.

46 ~~(O)~~ The maximum number of production rules

47 ~~(P)~~ CFG supports recursive non-phrase generation

48 ~~(Q)~~ structural (syntactic) ambiguity.

~~(R)~~ LALR ambiguity.

up) (b) the verb 'gave' requires additional complement

so) (b) vnp